



Polarization dynamics of vector solitons in a fiber laser

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Abstract: We investigate the polarization dynamics of vector solitons in a fiber laser mode-locked by a saturable absorber (SA). Three types of vector solitons were obtained in the laser, including group velocity locked vector solitons (GVLVS), polarization locked vector solitons (PLVS), and polarization rotation locked vector solitons (PRLVS). Their polarization evolution during intracavity propagation is discussed. Pure vector solitons are obtained from the continuous wave (CW) background by soliton distillation, and the characteristics of the vector solitons without and with distillation are analyzed, respectively. Numerical simulations suggest that the features of vector solitons in a fiber laser could be assemble to those generated in fibers.

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1. Introduction

Solitons are a specific kind of wave packet formed in nonlinear systems, which can maintain their shape and intensity during propagation. In optics, optical solitons usually refer to the optical field that remains constant during propagation due to the balance between dispersion and nonlinear effects in optical media [1]. In 1973, Hasegawa and Tappert theoretically predicted the existence of optical solitons in single-mode fibers (SMFs) [2,3], where the propagation of optical solitons is described by the nonlinear Schrödinger equation (NLSE). Ideally, SMFs are isotropic media. However, a practical fiber will always be weakly birefringent due to strains and bending, resulting in the different speeds of the two orthogonal polarization modes in the fiber. Owing to nonlinear effects, two polarization components can be coupled and propagate as a unit [4,5], which is called as the vector soliton. Vector solitons have been obtained in ultrafast fiber lasers applying various mode-locking techniques, such as semiconductor saturable absorber mirror (SESAM) [6], nonlinear amplifier loop mirror (NALM) [7], and saturable absorber (SA) [8]. In general, vector solitons cannot be formed in the cavity containing polarization discrimination devices. For this reason, numerical simulations or experiments of vector solitons will not employ the fiber laser mode-locked by nonlinear polarization rotation (NPR). With different cavity parameters and operation conditions, vector solitons present quite distinct polarization evolution, thus they can be classified as group velocity locked vector solitons (GVLVS) [6,9], polarization locked vector solitons (PLVS) [10,11], and polarization rotation locked vector solitons (PRLVS) [12,13].

Fiber lasers can easily control experimental parameters to generate solitons and have become an important platform for the investigation of optical solitons [14–16]. A fiber laser is a paradigm of dissipative systems, where solitons are affected by gain and loss while releasing dispersive

waves. Dispersive waves and solitons propagate in the cavity, and when the phase difference between them is multiples of 2π , Kelly sidebands are formed symmetrically on the spectrum due to the constructive interference [17]. At the same time, incoherent sidebands also appear in the spectrum [18]. These sidebands are known as continuous wave (CW) backgrounds combined with solitons. The incoherent sidebands can be removed by reducing the pump power or polarization modulation, but the coherent Kelly sidebands are difficult to be separated from the solitons. Therefore, the dynamics of pure solitons without the CW background are unclear.

Recently, a soliton distillation method based on nonlinear Fourier transform (NFT) has been proposed to remove the coherent sidebands [15,16]. The NFT originates from the inverse scattering transform (IST), which is widely used to solve nonlinear partial differential equations, such as the nonlinear Schrödinger equation (NLSE) [19,20]. The NFT can decompose the signal into soliton components (discrete spectrum) and non-soliton components (continuous spectrum). In the field of optical communication, NFT can effectively compensate for nonlinear impairments by transforming the time domain signal into the nonlinear domain, and the transmission performance beyond the limits of conventional Kerr nonlinearity is achieved [21]. In fiber lasers, NFT has been used to analyze laser radiation [22–25]. The nonlinear spectrum of the solitons generated by a mode-locked fiber laser can be obtained by NFT. In the nonlinear spectrum, the discrete spectrum contains the eigenvalues of the soliton and the sidebands, where the eigenvalues lie on the upper-half complex plane. The real parts of these eigenvalues correspond to the soliton frequency and the imaginary parts correspond to the soliton amplitude. Therefore, in the discrete spectrum, it is possible to distinguish and remove the sidebands from the soliton according to the different distributions of eigenvalues. Because the process of separating solitons and sidebands is similar to water distillation, this method is called as soliton distillation.

In this paper, we report the polarization dynamics of vector solitons in a passively mode-locked fiber laser. GVLVS, PLVS and PRLVS are obtained by varying parameters such as cavity birefringence and SA modulation depth. In addition, the three types of corresponding pure vector solitons are obtained by the NFT-based soliton distillation method, and the variation of their polarization states before and after distillation is discussed.

2. Theoretical model

In a lossless and noiseless fiber, the evolution of the slow-varying optical field envelope under the effects of dispersion and Kerr nonlinearity can be described by the normalized NLSE [26]:

$$i \frac{\partial q(t, z)}{\partial z} + \frac{1}{2} \frac{\partial^2 q(t, z)}{\partial t^2} + |q(t, z)|^2 q(t, z) = 0, \quad (1)$$

where $q(t, z)$ is the slow-varying optical field envelope in the anomalous dispersion region. z and t are the distance and time variation along the fiber, respectively. The NLSE belongs to a class of integrable nonlinear systems, which means that its initial conditions can be decomposed into nonlinear spectral data. For NLSE, this process is implemented by solving the Zakharov-Shabat problem [19]:

$$\frac{d}{dt} \begin{pmatrix} v_1(t, \zeta) \\ v_2(t, \zeta) \end{pmatrix} = \begin{pmatrix} -i\zeta & q(t) \\ -q^*(t) & i\zeta \end{pmatrix} \begin{pmatrix} v_1(t, \zeta) \\ v_2(t, \zeta) \end{pmatrix}, \quad (2)$$

where $q(t)$ is the signal with vanishing boundary. ζ is the spectral parameter (sometimes denoted as λ), which can be interpreted as a nonlinear analog of the frequency, and $*$ indicates the complex conjugate. The specific solutions to Eq. (2) are

$$\Phi^1(t, \zeta) = \begin{pmatrix} \phi_1^1 \\ \phi_2^1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\zeta t}, \quad t \rightarrow +\infty, \quad (3)$$

$$\Phi^2(t, \zeta) = \begin{pmatrix} \phi_1^2 \\ \phi_2^2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-i\zeta t}, t \rightarrow -\infty. \quad (4)$$

Then, the scattering coefficients can be expressed as

$$a(\zeta) = \lim_{t \rightarrow +\infty} \phi_1^2(t, \zeta) e^{i\zeta t}, \quad b(\zeta) = \lim_{t \rightarrow +\infty} \phi_2^2(t, \zeta) e^{-i\zeta t}. \quad (5)$$

And the nonlinear spectrum of $q(t)$ is defined as

$$\hat{q}_c(\zeta) = \frac{b(\zeta)}{a(\zeta)}, \quad \tilde{q}_d(\zeta_j) = \frac{b(\zeta_j)}{a_{\zeta}(\zeta_j)}, \quad \zeta_j \in C^+, \quad (6)$$

where $a(\zeta_j) = 0$, $a_{\zeta}(\zeta_j) = \frac{da(\zeta)}{d\zeta}|_{\zeta=\zeta_j}$. $\hat{q}_c(\zeta)$ is the continuous part of the nonlinear spectrum, which corresponds to the incoherent wave component of the signal. The discrete part (the coherent part) consists of the set of ζ_j and is called as the eigenvalue.

The inverse nonlinear Fourier transform (INFT) depends on the solution of the matrix Riemann-Hilbert problem [26]:

$$\begin{pmatrix} V^1(t, \zeta) & V^2(t, \zeta) \end{pmatrix} = \begin{pmatrix} \tilde{V}^1(t, \zeta^*) & \tilde{V}^2(t, \zeta^*) \end{pmatrix} \times \begin{pmatrix} \frac{b^*(\zeta^*)}{a^*(\zeta^*)} e^{-2j\zeta t} & \frac{1}{a^*(\zeta^*)} \\ \frac{1}{a^*(\zeta^*)} & \frac{b(\zeta)}{a^*(\zeta^*)} e^{2j\zeta t} \end{pmatrix}, \quad (7)$$

where $V^1(t, \zeta)$ and $V^2(t, \zeta)$ are canonical eigenvectors, $\sim$ represents taking the conjugate. And the signal $q(t)$ can be recovered from the nonlinear spectrum:

$$q^*(t) = 2i \sum_{j=1}^N \tilde{q}_d(\zeta_j) e^{2i\zeta_j t} V_2^1(t, \zeta_j) - \frac{1}{\pi} \int_{-\infty}^{\infty} \hat{q}_c(\zeta) e^{2i\zeta t} V_2^1(t, \zeta) d\zeta. \quad (8)$$

When there is no continuous spectrum and only one eigenvalue in the discrete spectrum, the corresponding signal $q(t)$ is a soliton. By applying the INFT on this eigenvalue, the soliton defined by the nonlinear spectral coefficients can be obtained:

$$q(t) = -2i\zeta_I e^{-i\angle \tilde{q}_d(\zeta_j)} \text{sech}(2\zeta_I(t - t_0)) e^{-2i\zeta_R t}, \quad (9)$$

where ζ_R and ζ_I are the real and imaginary parts of eigenvalue ζ_j . $t_0 = \frac{1}{2\zeta_I} \ln\left(\frac{|\tilde{q}_d(\zeta_j)|}{2\zeta_I}\right)$ is the time center of the soliton. $\angle \tilde{q}_d(\zeta_j)$ and $|\tilde{q}_d(\zeta_j)|$ are the phase and amplitude of the discrete spectrum, respectively. Therefore, a soliton can be fully characterized by the eigenvalue obtained by NFT, which is the theoretical foundation of our research.

The configuration of the mode-locked fiber laser used for numerical simulation is shown in Fig. 1. An erbium-doped fiber (EDF) of 2.6 m and the remaining SMFs with a total of 3 m constitute the main body of the ring cavity. In addition, an SA is employed to obtain mode-locked operation, ensuring there are no polarization discrimination devices in the cavity. The pulse trains are output by a 10:90 output coupler (OC). The group velocity dispersion (GVD) parameters for EDF and SMFs are both $\beta_2 = -23 \text{ ps}^2/\text{km}$, and the third-order dispersion (TOD) parameter is $\beta_3 = -0.13 \text{ ps}^3/\text{km}$. The nonlinear parameters of fibers are all $\gamma = 3 \text{ W}^{-1}\text{km}^{-1}$. We note that the parameters of the fiber laser and the following operation conditions are so selected that GVLVS, PLVS and PRLVS can be achieved in the fiber laser.

The pulse propagation in an SMF with birefringence could be described by coupled-mode equation [1], which is derived from the NLSE. However, unlike the SMF that can be considered as a conservative system if the propagation loss is ignored, the fiber laser is a typical dissipative system where pulses are affected by gain and loss as they are propagating. Therefore, the

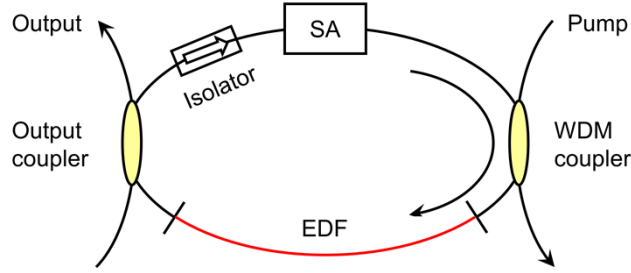


Fig. 1. Schematic of the fiber laser. WDM: Wavelength division multiplexer. EDF: Erbium-doped fiber. SA: Saturable absorber.

gain of the EDF needs to be taken into account in the equation to compliment the coupled Ginzburg-Landau equation (GLE)

$$\begin{aligned}
 \frac{\partial u}{\partial Z} = & j\frac{\Delta\beta}{2}u - \delta\frac{\partial u}{\partial T} - \beta_2\frac{j}{2}\frac{\partial^2 u}{\partial T^2} + \frac{\beta_3}{6}\frac{\partial^3 u}{\partial T^3} + j\frac{\gamma}{3}u^*v^2 + j\gamma\left(|u|^2 + \frac{2}{3}|v|^2\right)u \\
 & + \frac{g}{2}u + \frac{g}{2\Omega_g^2}\cdot\frac{\partial^2 u}{\partial T^2}, \\
 \frac{\partial v}{\partial Z} = & -j\frac{\Delta\beta}{2}v + \delta\frac{\partial v}{\partial T} - \beta_2\frac{j}{2}\frac{\partial^2 v}{\partial T^2} + \frac{\beta_3}{6}\frac{\partial^3 v}{\partial T^3} + j\frac{\gamma}{3}v^*u^2 + j\gamma\left(|v|^2 + \frac{2}{3}|u|^2\right)v \\
 & + \frac{g}{2}v + \frac{g}{2\Omega_g^2}\cdot\frac{\partial^2 v}{\partial T^2},
 \end{aligned} \tag{10}$$

where u and v denote the slowly varying envelopes of pulses polarized along the slow and fast axes of the fiber, respectively; $\Delta\beta = \beta_{0x} - \beta_{0y} = 2\pi/L_b$ is the difference in the wave number between two components, and L_b is the beat length of fibers; $2\delta = \beta_{1x} - \beta_{1y}$ is the group delay difference; β_2 and β_3 are the GVD and TOD parameters, respectively; γ is the nonlinear coefficient, which relates to the nonlinearity of the fiber. The penultimate term in Eq. (10) represents the gain provided by the EDF, while the last term denotes the gain dispersion. Here Ω_g refers to the gain bandwidth, and g is defined as

$$g = g_0 \exp \left[-\frac{1}{E_{\text{sat}}} \int_{-\infty}^t (|u|^2 + |v|^2) dt' \right], \tag{11}$$

where g_0 refers to the small-signal gain, and E_{sat} is the saturation energy, proportional to the pump power. In general, g_0 is set to 300 m^{-1} and E_{sat} is 100 pJ .

In addition, the laser is mode-locked by an SA, and its saturable absorption effect can be expressed as

$$T(I) = 1 - \frac{q}{1 + I/I_{\text{sat}}}. \tag{12}$$

Equation (12) is the transmission function of the SA, where q denotes the modulation depth, which is set to 0.15. I represents the instantaneous power of pulse through SA; I_{sat} is the saturation power and has been fixed at 200 W .

In the simulations, we adopted the Poincaré sphere to represent the polarization state of the pulse. For the different polarization states, the four Stokes parameters can be expressed as

$$\begin{aligned}
 S_0 &= |u|^2 + |v|^2, & S_1 &= |u|^2 - |v|^2, \\
 S_2 &= 2|u||v|\cos\Delta\varphi, & S_3 &= 2|u||v|\sin\Delta\varphi,
 \end{aligned} \tag{13}$$

where $\Delta\varphi$ denotes the phase difference between the components along the two orthogonal polarization directions. Since $S_0^2 = S_1^2 + S_2^2 + S_3^2$, S_1 , S_2 , and S_3 can represent a sphere of radius S_0 .

This sphere is called as the Poincaré sphere, and the polarization state of the pulse corresponds to a point on the sphere when S_0 is normalized to 1. As the polarization state varies, the Stokes vector consisting of these parameters moves on the sphere.

3. Polarization dynamics of vector solitons

Three types of vector solitons, GVLVS, PLVS, and PRLVS, were obtained by carefully adjusting the average linear birefringence L_b of the cavity and the modulation depth q .

3.1. Polarization dynamics of GVLVS

GVLVS is the most general type of vector soliton, with a wide range of parameters to choose from. When $L_b = L/20$, where L is the cavity length, GVLVS was obtained. As shown in Fig. 2(a), the peak intensities of two components are close and they are coupled together with almost overlapping pulse profiles. We note that the vertical axis corresponds to the fast axis of the fiber, while the horizontal axis corresponds to the slow axis. Figure 2(b) presents the spectra of the GVLVS and two components, where the GVLVS spectrum is artificially raised by 5 dB for ease of viewing when drawing the figure. It can be seen from the figure that the central wavelengths of the spectra of two components are clearly shifted at 1560 nm, and the sidebands are also shifted so that two sets of sidebands appear on the spectrum of GVLVS. Figure 2(c) shows the evolution of the phase difference between the components during intracavity propagation. The phase difference goes through a total of 40.0208π from 0.0035π to 0.0243π , where 40π is contributed by linear birefringence and 0.0208π comes from nonlinear birefringence induced by nonlinearity. We plot the Stokes vector for every 10 cm propagation in the fiber laser, as shown in Fig. 2(d). With GVLVS propagating in the cavity, the Stokes vector moves on the Poincaré sphere, indicating that the polarization state of the vector soliton is varying all the time. Figure 2(e) displays the polarization evolution of GVLVS at the position of OC. Since the polarization rotation is always not multiples of 2π after each round trip in the cavity, the polarization state of the output soliton will keep changing, and the Stokes vector circles around the Poincaré sphere. The peak intensity evolution of the output GVLVS without and with an external polarizer is given in Fig. 2(f), where the transmission axis of the polarizer is fixed at an angle of 45° to the horizontal axis. Without the external polarizer, the output solitons are stable and the peak intensity varies slightly. However, when solitons pass through the polarizer, the peak intensity varies continuously, suggesting that the polarization states of solitons are also varying.

The stable state of GVLVS is attributed to the dynamic balance of birefringence and cross-phase modulation (XPM). Birefringence leads to the different group velocities of two orthogonal direction components. However, in the anomalous dispersion region, the XPM effect shifts the fast and slow axis components to the long and short wavelength directions, respectively. The corresponding frequency difference compensates for the inconsistent group velocity, allowing the two components to propagate together. For each round trip of the soliton in the cavity, the nonlinear birefringence results in a polarization rotation that is not an integer number of turns, and hence the output soliton polarization state is constantly varying.

3.2. Polarization dynamics of PLVS

When the cavity birefringence was very weak with $L_b = 1000L$, the linear birefringence was comparable to the nonlinear birefringence and PLVS could be obtained. To obtain a stable state, the small-signal gain g_0 should be kept between 290 m^{-1} and 310 m^{-1} . As shown in Fig. 3(a), the profiles of the two components almost overlap, since the cavity is approximately an isotropic medium. The corresponding spectra in Fig. 3(b) also nearly overlap. The evolution of phase difference between two components is presented in Fig. 3(c), the phase difference is fixed at $\pi/2$, indicating that the vector soliton is locked in circular polarization. Figure 3(d) shows the Stokes vector for every 10 cm propagation in the fiber laser on the Poincaré sphere. There is

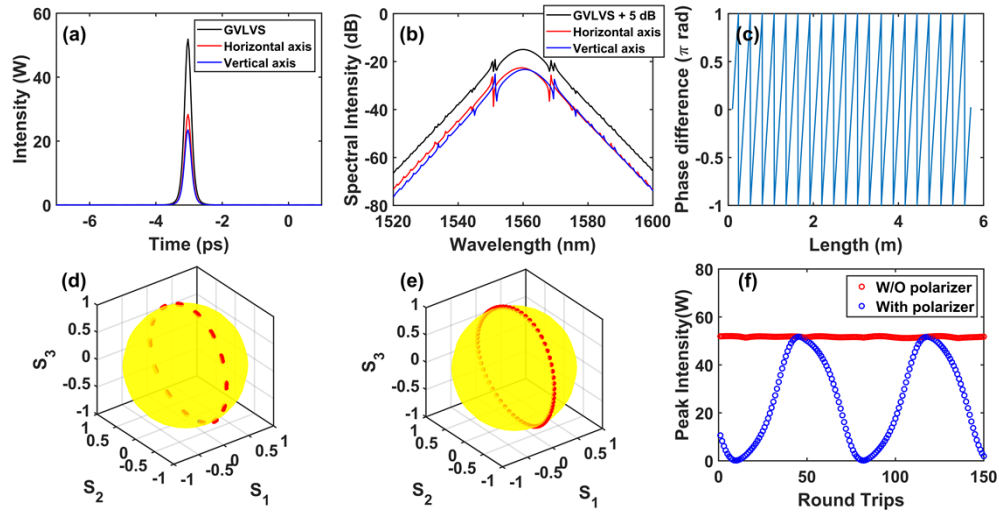


Fig. 2. Properties of GVLVS. (a) Temporal profiles of GVLVS and its two polarization components from the OC. (b) Optical spectra of them. (c) Evolution of the phase difference between two polarization components in one round trip. (d) Polarization evolution of GVLVS shown on the Poincaré sphere in one round trip. (e) Polarization evolution of the output GVLVS. (f) Peak intensity evolution of the output GVLVS without and with the external polarizer.

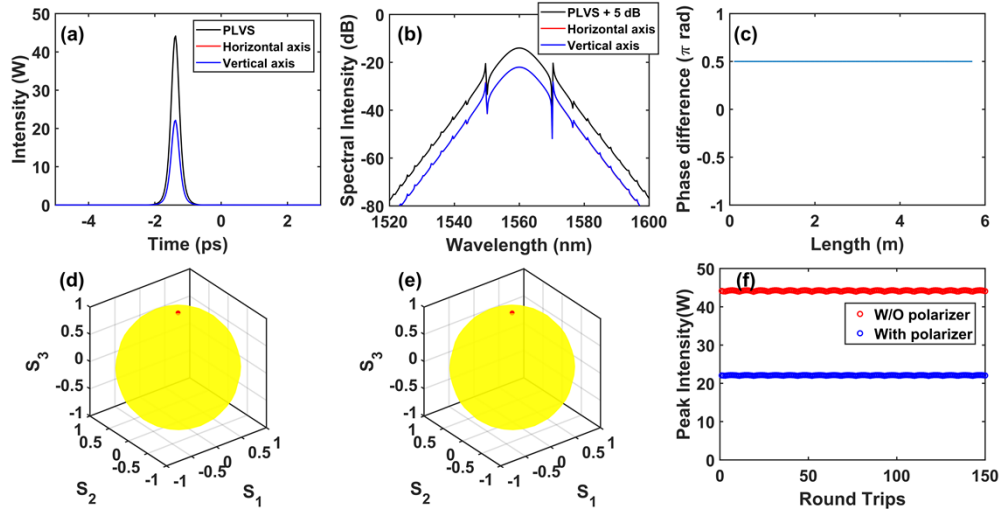


Fig. 3. Properties of PLVS. (a) Temporal profiles of PLVS and its two polarization components from the OC. (b) Optical spectra of them. (c) Evolution of the phase difference between two polarization components in one round trip. (d) Polarization evolution of PLVS shown on the Poincaré sphere in one round trip. (e) Polarization evolution of the output PLVS. (f) Peak intensity evolution of the output PLVS without and with the external polarizer.

only one fixed point on the surface in one round trip. Moreover, Fig. 3(e) shows that there is also only one fixed point on the Poincaré sphere of the output PLVS. Finally, as illustrated in Fig. 3(f), after passing through a polarizer that remains at an angle of 45° to the horizontal axis,

the peak intensity of the output PLVS is exactly half of that without the polarizer, thus the PLVS is circularly polarized.

For PLVS with low birefringence, there is a slight difference in the peak power between the two components. For example, the peak power of the vertical axis (fast axis) in Fig. 3(a) is 0.113 W higher than that of the horizontal axis (slow axis), so the fast axis gains a larger nonlinear refractive index than the slow axis due to the XPM. Therefore, the additional nonlinear refractive index compensates for the difference in refractive index between the two axes, allowing both components to propagate at the same phase velocity with constant polarization.

3.3. Polarization dynamics of PRLVS

When the beat length is approximately 0.2 to 2 times the cavity length and the small-signal gain g_0 is kept at 290 m^{-1} to 310 m^{-1} , PRLVS can be obtained easily by trimming the q . In the simulations, when $L_b = 2L$ and $q = 0.3$, as shown in Figs. 4(a) and 4(b), the time profiles and the central wavelengths of the spectra of the two components almost overlap, and the peak-dip relationship appears in spectra due to coherent energy exchange [13]. Figure 4(c) shows the evolution of the phase difference $q = 0.3$ between components in the cavity. In one round trip, the phase difference increases approximately linearly from -0.1π to 0.9π . Therefore, the variation of the phase difference is 2π after two round trips. The Stokes vector on the Poincaré sphere in two

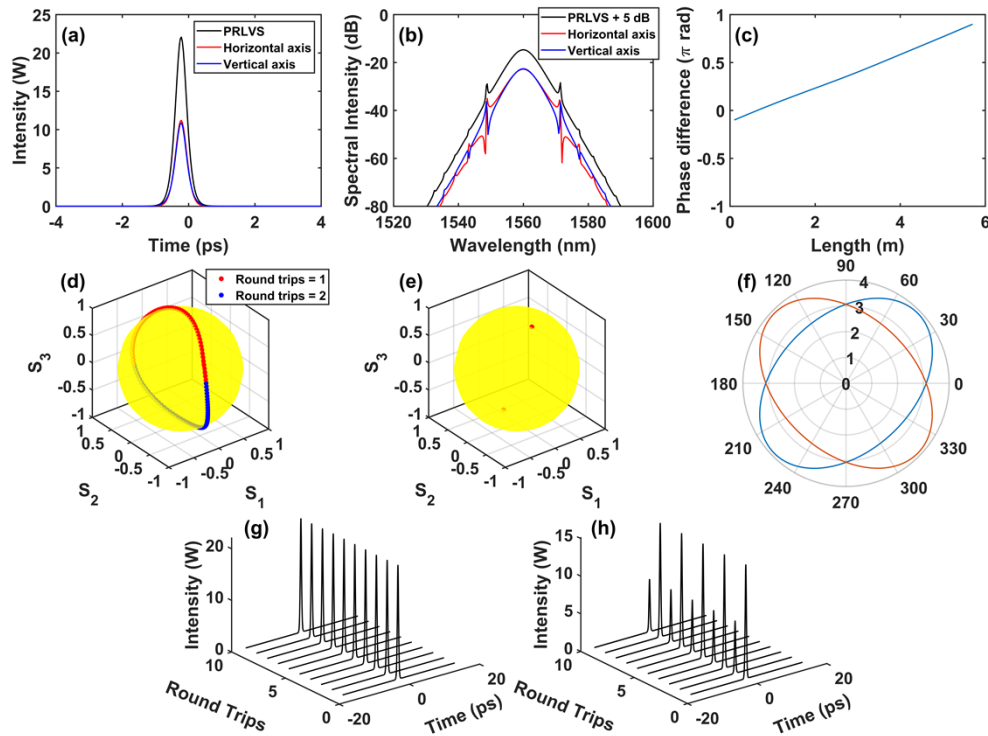


Fig. 4. Properties of PRLVS. (a) Temporal profiles of PRLVS and its two polarization components from the OC. (b) Optical spectra of them. (c) Evolution of the phase difference between two polarization components in one round trip. (d) Polarization evolution of PRLVS shown on the Poincaré sphere in two round trips. (e) Polarization evolution of the output PRLVS. (f) Corresponding polarization ellipses. (g) Time profile evolution of PRLVS from OC without the external polarizer. (h) Time profile evolution of PRLVS from OC with the external polarizer.

round trips is shown in Fig. 4(d), the evolution step is still every 10 cm propagation in the cavity. The Stokes vector in a circle indicates that the polarization state of PRLVS returns to the original state with every two round trips. As shown in Fig. 4(e), the two points on the sphere correspond to the two polarization states of the output PRLVS, and they differ from each other by π . The corresponding polarization ellipses are shown in Fig. 4(f). Figures 4(g) and 4(h) present the time profile evolution of PRLVS from OC without and with the external polarizer, respectively. The transmission axis is also set at an angle of 45° to the horizontal axis. In Fig. 4(g), the output pulses are stable, but the pulses go into the state of period doubling after passing through the external polarizer (Fig. 4(h)).

The formation of PRLVS is attributed to the effect of the cavity boundary condition. Due to the freedom of polarization evolution inside the cavity, the polarization evolution frequency [27] is inversely proportional to the birefringence beat length. Consequently, if the cavity length is close to multiple times of the birefringence beat length, the cavity boundary condition will apply an extra periodic force on the pulse polarization evolution. In another word, the soliton polarization rotation can be locked by the cavity boundary condition when the period of the soliton polarization rotation is close to the multiple of the cavity roundtrip time [12]. Figure 4 shows the case when the period of the soliton polarization rotation is twice of the cavity roundtrip time.

Similar to GVLVS, the polarization state of PRLVS also varies during the propagation, but the number of round trips is equal to the ratio of beat length to cavity length when the polarization state returns to its original state. For example, in this simulation, the ratio of beat length to cavity length is 2, then the polarization state of PRLVS returns to itself after two round trips, and the phase difference increases by 2π .

4. Polarization dynamics after soliton distillation

Before applying NFT to analyze a laser pulse, it is important to choose a proper form of normalization. Different normalization parameters can lead to different energy distributions in the nonlinear spectrum, and the number of eigenvalues also varies [25]. In this paper, we adopt a simple approach by scaling the pulse amplitude for the sake of simplification:

$$q'(t, z) = q(t, z)/A, \quad (14)$$

where A is a real number. The energy distribution and the number of eigenvalues change when the value of A is varied. Typically, a suitable value for A is chosen such that the maximum energy is allocated to the discrete spectrum.

Following the calculation procedure in Ref. [15,16], the continuous and discrete spectrum of the GVLVS components along two orthogonal polarization directions are shown in Fig. 5. The continuous spectrum is shown in Fig. 5(a), which represents the dispersive component of the soliton. There exists slight difference between the background along h-axis and the one along v-axis. As the GVLVS components share gain with background, there are a lot of factors affecting the difference between the background along h-axis and the one along v-axis. For example, the intensity, phase, and central wavelength of the two components of GVLVS may contribute to the difference. Figure 5(b) shows the eigenvalues of the two components along the h-axis and the v-axis. The distributions of the two sets of eigenvalues are quite different. The real parts of the h-axis eigenvalues are larger than that of the v-axis eigenvalues, and the corresponding imaginary parts have little difference. This indicates that the central wavelengths of the two soliton components are clearly shifted, but the intensity difference is small, which can be corroborated by Figs. 2(a) and 2(b). In addition, we can easily identify that the eigenvalues with larger imaginary parts correspond to solitons, and those with smaller imaginary parts on both sides correspond to sidebands because the intensity of solitons is generally much larger than

that of sidebands. After that, the eigenvalues with smaller imaginary parts could be removed, and then solitons without sidebands can be reconstructed by applying INFT.

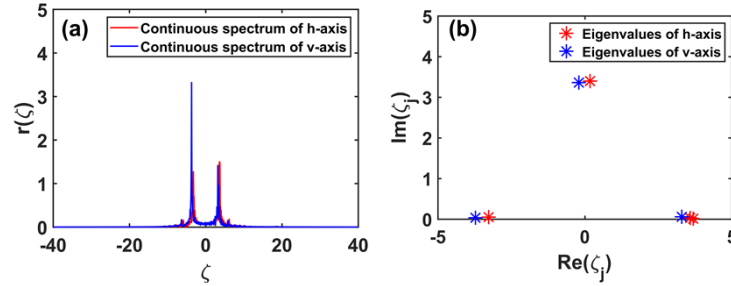


Fig. 5. The (a) continuous spectrum and (b) discrete spectrum of GVLVS. The h-axis refers to the horizontal axis, and the v-axis refers to the vertical axis. A_h (A of h-axis) is 1.36 and A_v (A of v-axis) is 1.43.

As shown in Fig. 6(a), after soliton distillation, the peak intensity of the soliton decreases because the CW background is removed. The variation of the spectrum is shown in Fig. 6(b), the paired Kelly sidebands on the spectrum have been eliminated. Moreover, the polarization ellipse of GVLVS shown in Fig. 6(c) also changes, the ellipse gets smaller. However, the major axis orientation and flatness of the ellipse change slightly since the CW background contains negligible energy compared to the solitons.

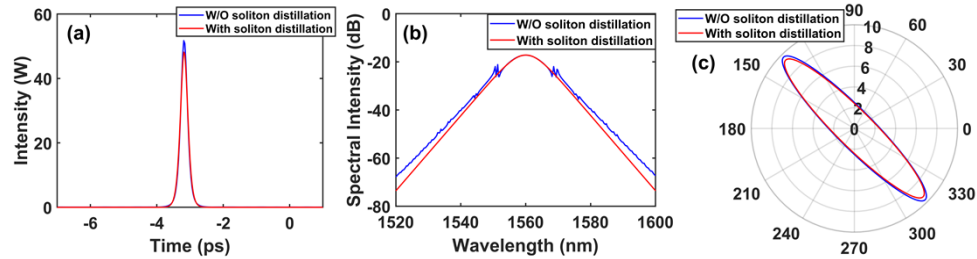


Fig. 6. (a) Temporal profiles, (b) spectra, and (c) polarization ellipses of GVLVS without and with soliton distillation.

Furthermore, PLVS can also be removed the CW background by using soliton distillation. Figures 7(a) and 7(b) show the temporal profiles and the corresponding spectra. It can be seen that the extra energy in the background and the Kelly sidebands are removed. As illustrated in Fig. 7(c), the polarization state of PLVS is almost constant without and with soliton distillation and is circularly polarized.

Similarly, the simulations of PRLVS after soliton distillation are shown in Fig. 8. As we set the modulation depth of SA to $q = 0.3$ in the simulations, the stronger saturable absorption effect weakens the CW background energy. As a result, both the variation of peak intensity and the sidebands shown in Fig. 8(a) and 8(b) are much smaller comparing to the cases of GVLVS and PLVS, and the polarization ellipses without and with soliton distillation in Fig. 8(c) are almost unchanged.

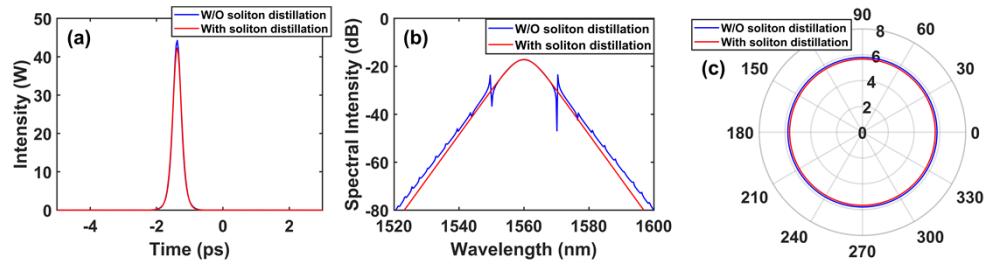


Fig. 7. (a) Temporal profiles, (b) spectra, and (c) polarization ellipses of PLVS without and with soliton distillation. A_h and A_v are both 1.25.

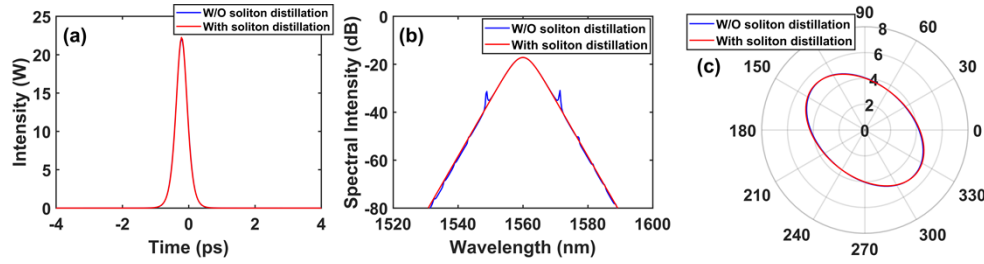


Fig. 8. (a) Temporal profiles, (b) spectra, and (c) polarization ellipses of PRLVS without and with soliton distillation. A_h is 1.35 and A_v is 1.32.

5. Discussion

The application of NFT requires the full field information of the pulse, which requests the support of coherent detection [25,28]. We note that to ensure the pulse without distortion during coherent detection, the digitally photo-detecting bandwidth should cover the pulse bandwidth. In practice, to demonstrate our proposed theme in experiments, a fiber laser with output pulse duration that can be covered by the photo-detecting bandwidth should be specially designed and exploited.

The equivalent cavity birefringence is a key factor determining the different regimes (GVLVS, PLVS, PRLVS). The equivalent cavity birefringence is determined by the fiber segments and the setting of the polarization controller inside the cavity. Large birefringence is easy to be set if we use a high birefringence fiber (or a polarization maintaining fiber). For small birefringence, apart from using a weak-birefringence fiber, the application of a polarization controller is critical since we can modify the equivalent cavity birefringence by tuning the polarization controller. A cavity with near zero equivalent birefringence may be achieved with carefully tuning the polarization controller.

6. Conclusion

In conclusion, we have reported polarization dynamics of vector solitons formed in a fiber laser mode-locked by an SA. We have obtained GVLVS, PLVS, and PRLVS under different linear cavity birefringence conditions. At most parameter settings, GVLVS is easily obtained but PRLVS is formed by carefully adjusting the parameters. In addition, PLVS is obtained only when the birefringence is very weak. The polarization evolution of these three types of vector solitons in cavity have been analyzed on the Poincaré sphere. Moreover, we used NFT-based soliton distillation to distill pure vector solitons from CW background. The polarization dynamics of three different vector solitons without/with soliton distillation are discussed. Our simulations pave a way to get insight into the study of pure vector solitons in fiber lasers.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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